

Is it decidable if a TM M accepts ϵ ?

Is $\{M \mid \epsilon \text{ is in } L(M)\}$ recursive?

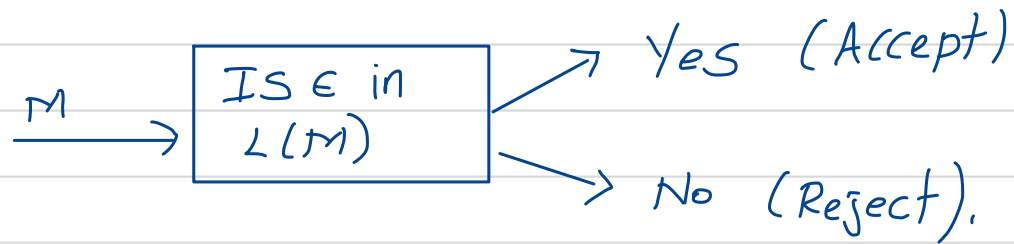
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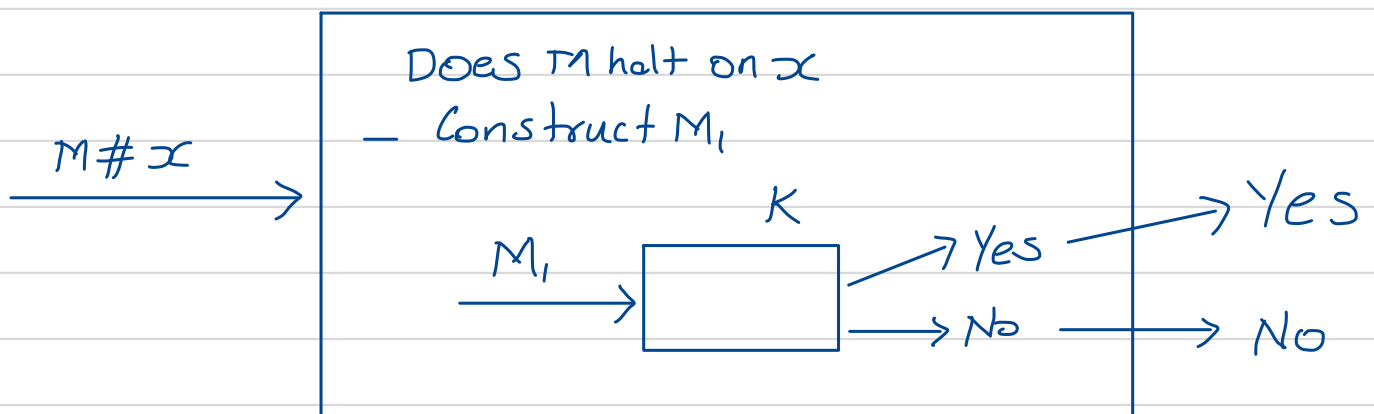
Suppose $\exists$ a total TM K that can decide if a given TM M accepts ϵ . i.e. $L(K) = \{M \mid \epsilon \text{ is in } L(M)\}$

Then we can construct a total TM K' s.t. $L(K') = HP$

A total TM K



A total TM K'



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- Given a TM M and x , to determine if M halts on x .

Construct M_1 that on input y works as follows.

1. Erases its input y .
 2. writes x on the tape.
 3. Runs M on input x
- } M and x are hardcoded in M' .

4. Accept if M halts on x .

if M halts on x , M_1 accepts y for all input y .

Thus $L(M_1) = \begin{cases} \Sigma^* & \text{if } M \text{ halts on } x \\ \emptyset & \text{if } M \text{ does not halt on } x. \end{cases}$

Run K with input M_1 ,

if K accepts $\Rightarrow \epsilon \in L(M_1) \Rightarrow L(M_1) = \Sigma^* \Rightarrow M$ halts on x

if K rejects $\Rightarrow \epsilon \notin L(M_1) \Rightarrow L(M_1) = \emptyset \Rightarrow M$ does not halt on x .

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Questions.

- Is it decidable if for a TM M , $L(M) = \Sigma^*$?

Is $\{M \mid L(M) = \Sigma^*\}$ recursive?

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Questions.

- Is it decidable if for a TM M , $L(M) = \emptyset$?
Is $\{M \mid L(M) = \emptyset\}$ recursive?

- Is it decidable if for a TM M , $L(M) = \Sigma^*$?
Is $\{M \mid L(M) = \Sigma^*\}$ recursive?

Same construction as above.

Question. Is it decidable, for a TM M , if $L(M)$ is regular.
Is $\{M \mid L(M) \text{ is regular}\}$ recursive?

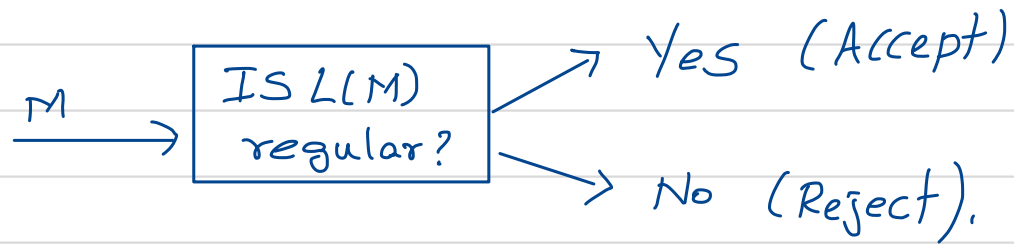
Choose a set A that is r.e. but not recursive.

Eg. $A = HP$ or $A = MP$. Let N be a TM s.t. $L(N) = A$.

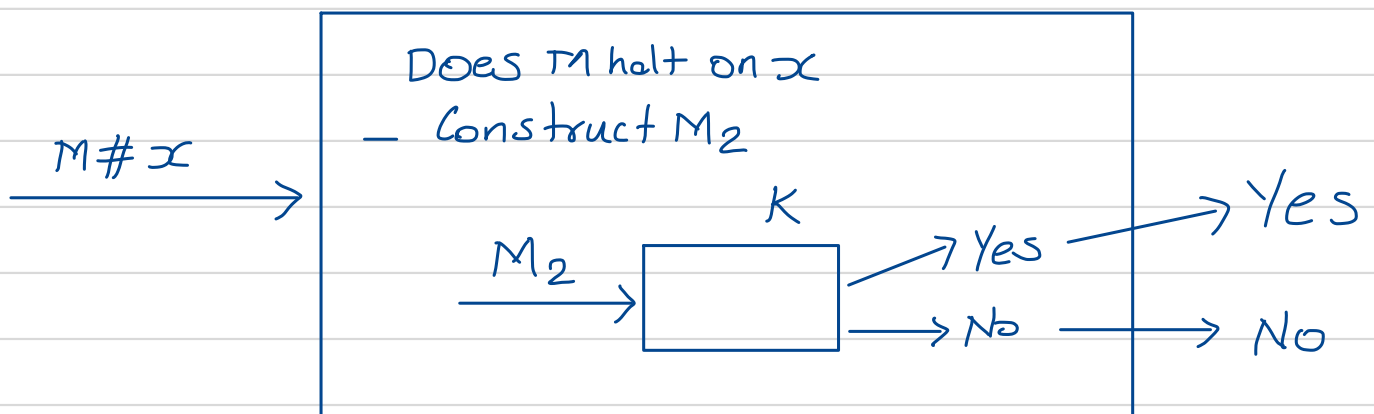
Suppose $\exists$ a total TM K that can decide, given an arbitrary TM M if $L(M)$ is regular.

Then using K , we can construct a total TM K' s.t.
 $L(K') = HP$

A total TM K



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Question. Is it decidable, for a TM M , if $L(M)$ is not regular
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 $L(K') = HP$

Given M and x to determine if M halts on x .

Construct a TM M_2 which on input y does the following:
 $\hookrightarrow$ with multiple tracks.

1. Writes y on one of the tracks
2. Writes x on a separate track.
3. Runs M on input x .

$\left. \begin{array}{l} M \text{ and } x \text{ are} \\ \text{hard coded in } M_2 \end{array} \right\}$

4. if M halts on x then M_2 runs N on y .
(y is M_2 original input)
 M_2 accepts if N accepts y .

if M does not halt on x then M_2 does not accept any string

$$L(M_2) = \begin{cases} A & \text{if } M \text{ halts on } x. \\ \emptyset & \text{if } M \text{ does not halt on } x. \end{cases}$$

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A is not recursive, so A is not regular; not a CFL
 $\emptyset$ is regular, CFL and recursive.

Techniques to show undecidability:

Diagonalization and Reduction.

To show problem B is undecidable - reduce HP to B .

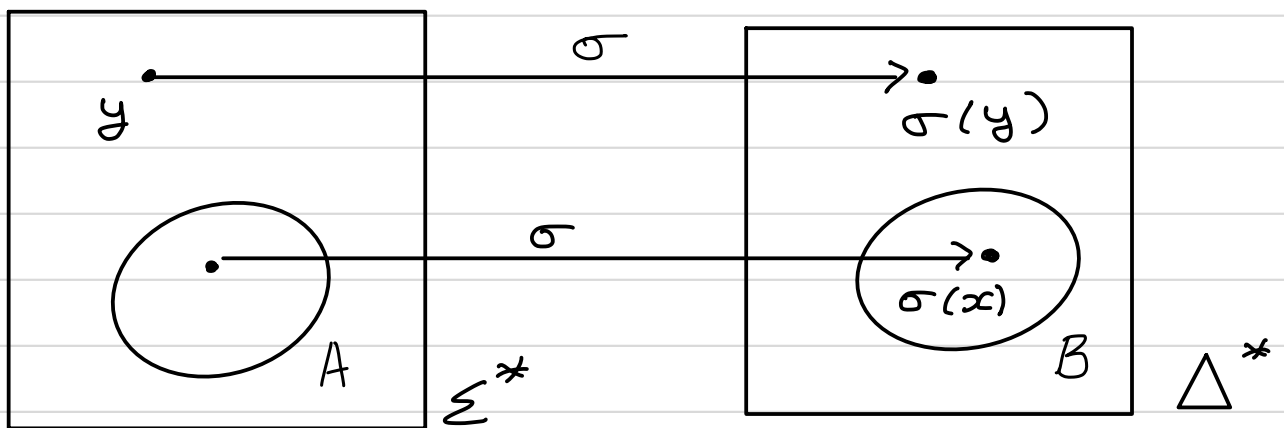
That is, Yes instances of HP become Yes instances of B .
No instances of HP become No instances of B .

IF B is decidable, use the decision procedure of B as a subroutine to get a decision procedure for HP - Contradiction.

Reduction.

Given $A \subseteq \Sigma^*$, $B \subseteq \Delta^*$, a (many-one) reduction of A to B is a computable function

$\sigma: \Sigma^* \rightarrow \Delta^*$ s.t. $\forall x \in \Sigma^*$, $x \in A$ iff $\sigma(x) \in B$



σ should be computable by a total TM.

A total TM that on any input x writes $\sigma(x)$ on the tape and halts.

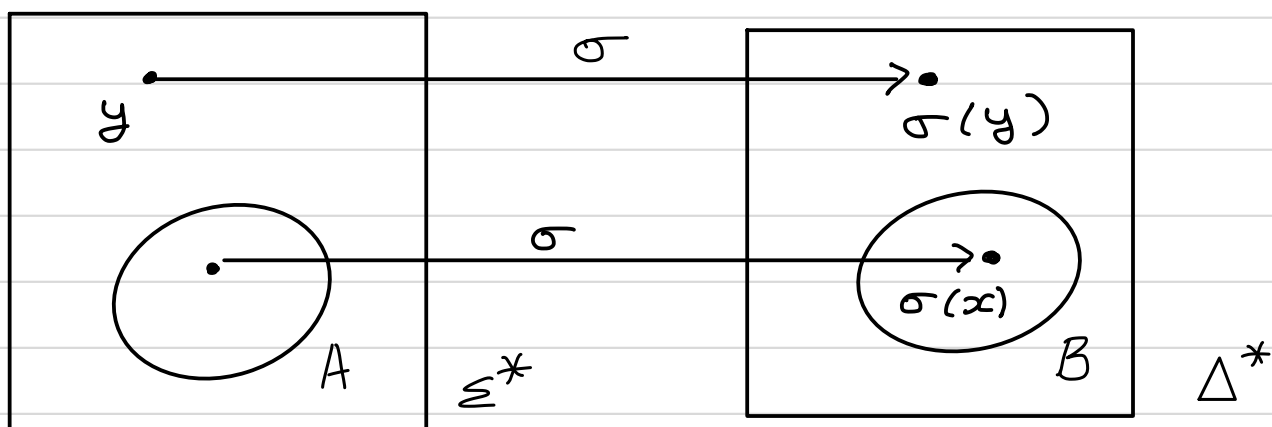
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Observations.

- $\leq_m$ between sets is transitive.

- if $A \leq_m B$ via σ then $\overline{A} \leq_m \overline{B}$ via σ

Theorem.

1. if $A \leq_m B$ and B is r.e then A is r.e.

↳ if $A \leq_m B$ and A is not r.e then B is not r.e.

2. if $A \leq_m B$ and B is recursive then A is recursive

↳ if $A \leq_m B$ and A is not recursive then B is not recursive.